

image handle or simply a pointer to the image. The parameters in *ImageCreate()* are the width and height of our rectangle.

Using the function *ImageColorAllocate()* with four parameters, the colour can be set to the image. The first parameter is the handle or pointer, and the next three parameters determine the colour. The colours are red, green and blue and must be an integer between 0 and 255. After this step, select the text colour, using the same format as our background colour. In this example, black is selected. Next, the text that is to be printed is entered.

The function *ImageString()* performs this step with multiple parameters.

- The first parameter is the image pointer.
- Then it's the font, and starting the X and Y co-ordinates.
- Next comes the text to be displayed, and last of all, the colour.

Finally, *ImagePng()* creates the dynamic PNG image.

```
<?php
header ("Content-type: image/png");
$imagepointer = ImageCreate (130, 50) or die ("Cannot Create image");
$bg_color = ImageColorAllocate ($imagepointer, 255, 0, 0);
$txt_color = ImageColorAllocate ($imagepointer, 0, 0, 0);
ImageString ($imagepointer, 10, 15, 28, "Good Morning",
$txt_color);
ImagePng ($imagepointer);
?>
```

To manipulate with fonts, type:

```
<?php
header ("Content-type: image/png");
$imagepointer = ImageCreate (130, 50) or die ("Cannot Create image");
$bg_color = ImageColorAllocate ($imagepointer, 255, 0, 0);
$txt_color = ImageColorAllocate ($imagepointer, 0, 0, 0);
ImageTTFText ($imagepointer, 20, 15, 30, 40, $txt_color, "/
Fonts/MyFont.ttf", "MyFont ");
ImagePng ($imagepointer);
?>
```

Using the *ImageTTFText()* function, programmers can use their own fonts that are available in the TTF format. In this function, there are multiple parameters. These are: image pointer, font size, rotation, X and Y co-ordinates, text colour, the font and text to be displayed, etc. In the font parameter, you need to include the path to the font file. For this example, the font *MyFont* is used in a folder called *Fonts*. As in the above cited code, the text will be displayed at a 15° angle. If the text isn't displayed, it means that the path to the font is wrong or the parameters of rotation, as well as the X and Y co-ordinates, are placing the text outside

the viewable area.

The following example outputs a 3D pie chart (from the PHP GD documentation on the *imagefilledarc()* function).

```
<?php
// Create an image
$image = imagecreatetruecolor(100, 100);

// Allocate some colors
$white = imagecolorallocate($image, 0xFF, 0xFF, 0xFF);
$gray = imagecolorallocate($image, 0xC0, 0xC0, 0xC0);
$darkgray = imagecolorallocate($image, 0x90, 0x90, 0x90);
$navy = imagecolorallocate($image, 0x00, 0x00, 0x80);
$darknavy = imagecolorallocate($image, 0x00, 0x00, 0x50);
$red = imagecolorallocate($image, 0xFF, 0x00, 0x00);
$darkred = imagecolorallocate($image, 0x90, 0x00, 0x00);

// Make the 3D effect
for ($i = 60; $i > 50; $i--) {
    imagefilledarc($image, 50, $i, 100, 50, 0, 45,
$darknavy, IMG_ARC_PIE);
    imagefilledarc($image, 50, $i, 100, 50, 45, 75,
$darkgray, IMG_ARC_PIE);
    imagefilledarc($image, 50, $i, 100, 50, 75, 360,
$darkred, IMG_ARC_PIE);
}

    imagefilledarc($image, 50, 50, 100, 50, 0, 45, $navy,
IMG_ARC_PIE);
    imagefilledarc($image, 50, 50, 100, 50, 45, 75, $gray,
IMG_ARC_PIE);
    imagefilledarc($image, 50, 50, 100, 50, 75, 360, $red,
IMG_ARC_PIE);

// Flush the image
header('Content-type: image/png');
imagepng($image);
imagedestroy($image);
?>
```

Graphics and chart libraries for PHP

Nowadays, a number of graphics and charting libraries are available that allow programmers to integrate charts and dynamic maps without any GD programming. Programmers make use of simple APIs to integrate the graphical reports and charts in the Web applications. This method saves lots of time in specialised GD programming. Third party APIs and libraries provide the code snippets with their libraries to enable Web programmers to rapidly develop Web applications.

Today, Web developers use assorted free chart libraries to display visualisations on websites. The following dynamic charting and graphics libraries are currently being used by PHP developers: